FORTINET | USA
XPERTS26

AWS 301: Extending Global SD-WAN into AWS Cloud WAN

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AWS 301 · FORTINET SECURE SD-WAN

FortiGate SD-WAN — topology + takeaways on one page

WHERE IT'S DEPLOYED

One engine, every topology

01 DIA at the branch

Single FortiGate, multi-ISP. Per-flow steering with performance SLAs; mid-session failover without user impact.

SD-WAN Underlay & App Monitoring sets link speeds dynamically.

02 Site-to-site + ADVPN

IPsec/GRE overlays over any underlay. App-aware per-path (voice→MPLS, bulk→internet) with iBGP across the overlay.

ADVPN shortcuts cut hub hairpin — spokes talk directly after broker.

03 Cloud on-ramp

Overlays from branches to FortiGate VMs in-cloud or native gateways; SLA-driven path selection to the best POP.

AWS: TGW / CWAN Connect, VPN, or Direct Connect attach.

04 SASE + remote breakout

FortiSASE joins the fabric as a spoke; roaming FortiClient users get full inspection at the POP, not the DC.

ADVPN shortcuts into FortiSASE POPs — no hairpin for user→branch.

WHY IT WINS

Four differentiators

01 One platform, not a stack

Routing + NGFW + SD-WAN + segmentation converge in one FortiOS appliance.

02 Secure by design

Full inspection (IPS, URL, app-control) runs natively at the branch, not downstream.

03 Every path, same engine

DIA, site-to-site, cloud on-ramp, SASE — same SD-WAN engine and policy model.

04 Self-optimizing steering

SLAs pick the best link in real time; ADVPN brokers direct spoke-to-spoke tunnels.

AWS tie-in FortiGate attaches via TGW, Cloud WAN (Connect / tunnel-less), VPN, or Direct Connect — same SD-WAN policy, now in the AWS fabric.

Agenda



Why Cloud WAN?



AWS Cloud WAN Intro



AWS Cloud WAN Components



Cloud WAN Routing



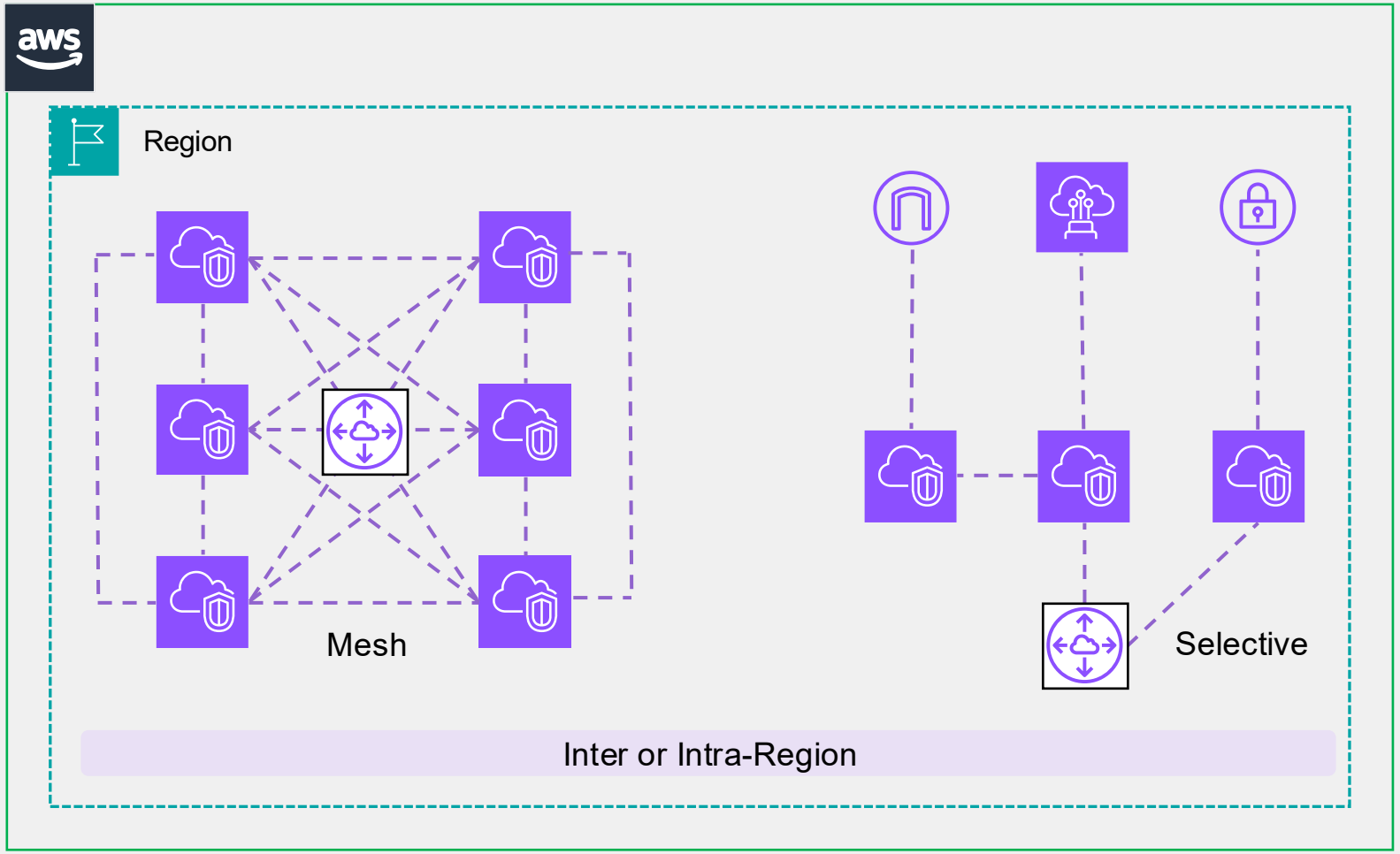
Cloud WAN Use Cases

AWS VPC Network Connectivity Options

VPC Peering

Key design considerations:

- Selective vs Mesh (Compulsion)
- Routing Control
  - Lack of Centralised routing control
  - No transitive routing supported
- Scalability and Complexity
  - Default 25 and max 125 peering connections per VPC
- Cost 0.098 USD per GB (intra region, higher for inter-region)



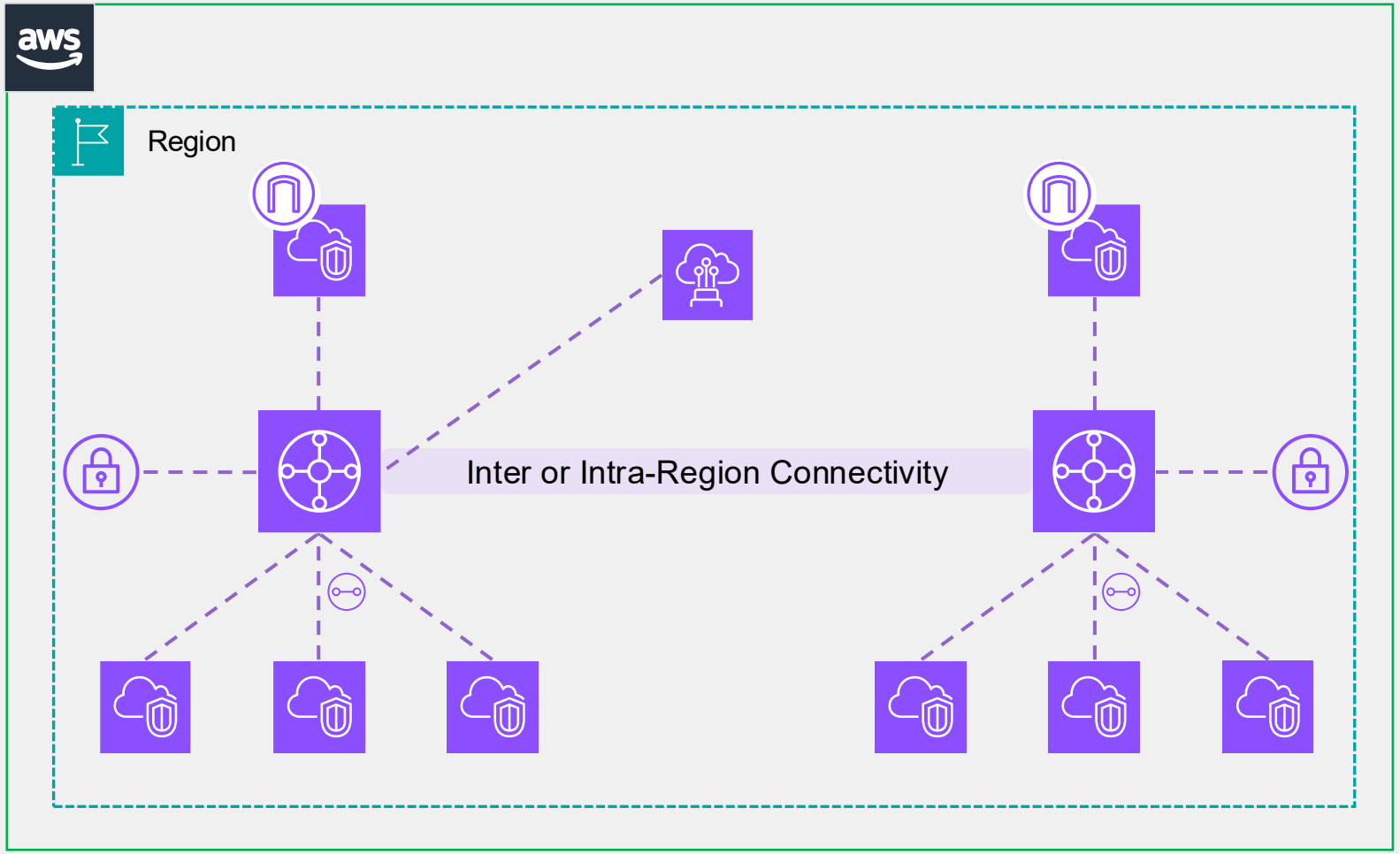
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AWS VPC Network Connectivity Options

TGW - Regional Construct

Key design considerations:

- Inter and Intra Region support
  - Jumbo MTU support
- Selective vs Mesh design
  - Transit Routing support
- Routing Control
  - Centralised at TGW layer per region
- Scalablity and Complexity
  - Tiered Hub and Spoke Topology
  - Default 5 TGW adjustable.
  - Default 5000 attachment
  - Default 50 Peering attachment.
- Cost 0.02 USD per GB per region.
  Additional TGW+Attachment per hour cost.

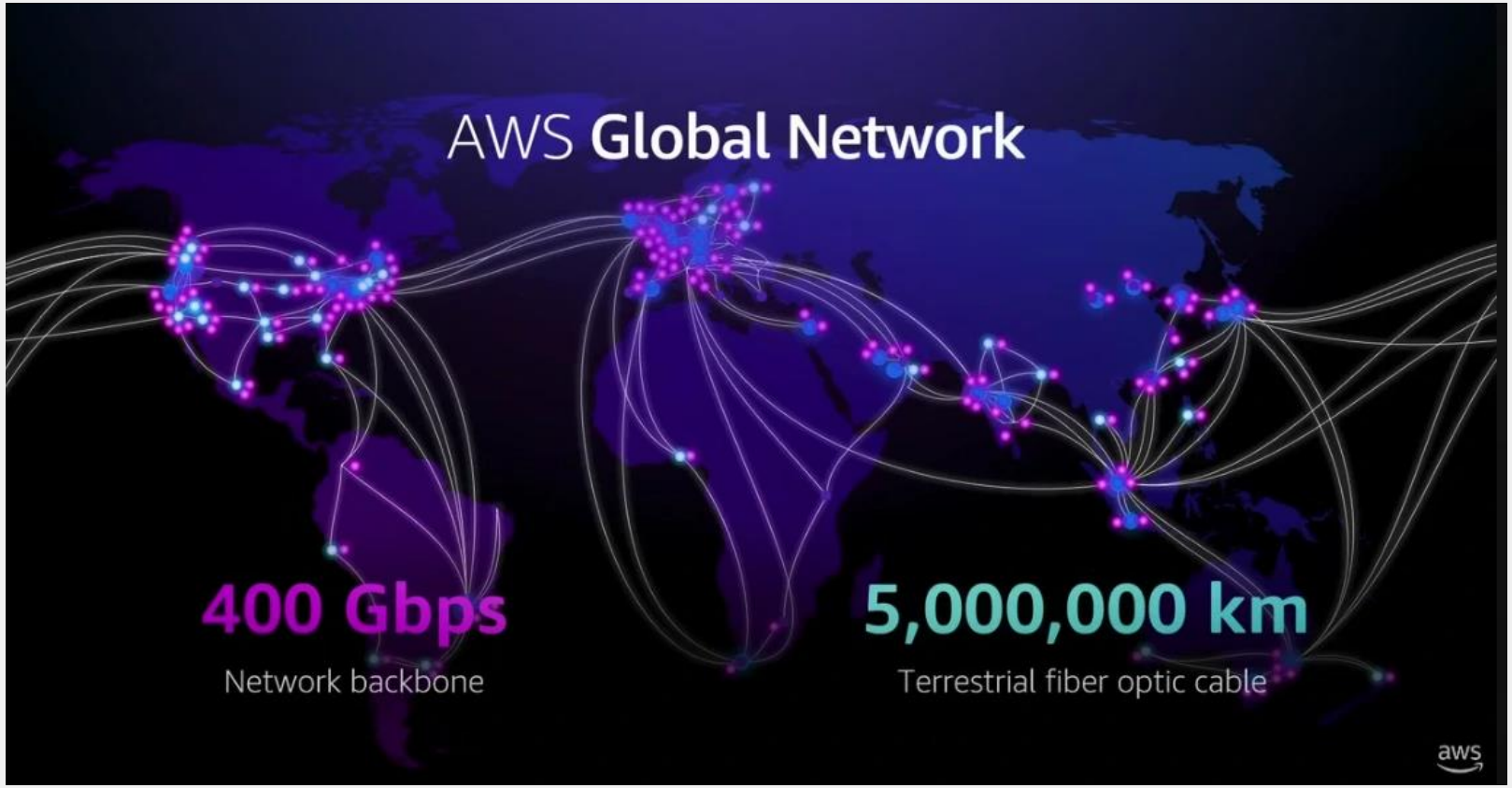


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AWS Cloud WAN

Global WAN Network

- AWS backbone at Core
- Inter and Intra Region support
  - Jumbo MTU support
- Bus Topology
  - Transit Routing support
- Routing Control
  - Centralised at Cloud WAN
- Scalable
  - Tiered Hub and Spoke with Bus Topology
  - Default 5 Global Networks.
  - Default 40 Segments
  - Default 500 attachment.
  - Default 50 TGW
- Cost 0.02 USD per GB per region
  Additional Attachment and CNE per hour cost.

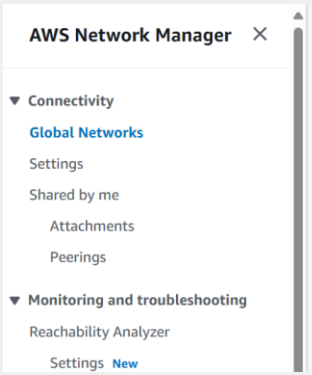


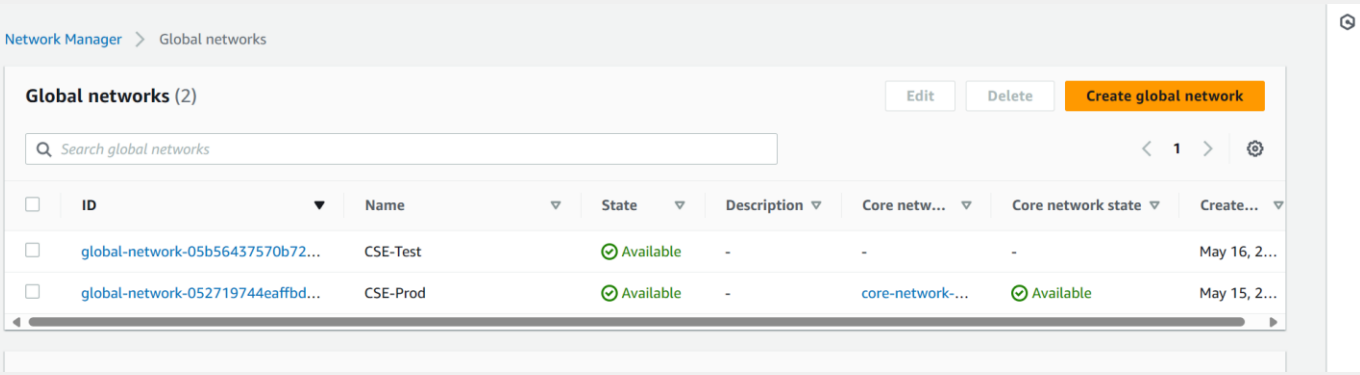
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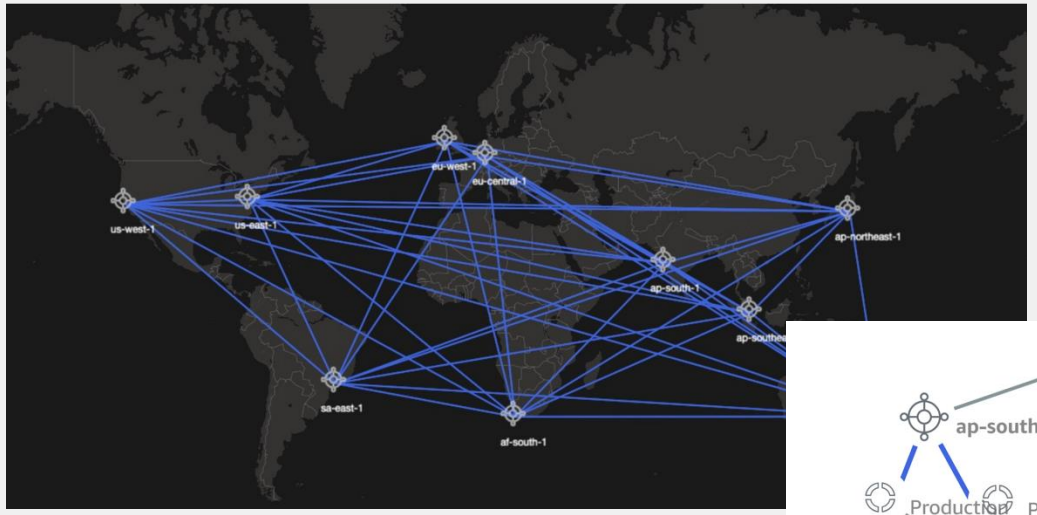
AWS Cloud WAN

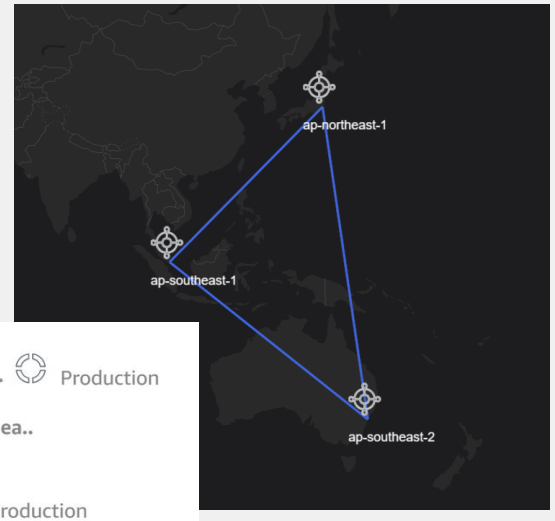
Key Components:

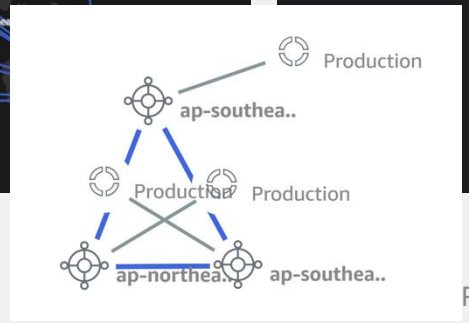
- Global Network Container
  - Core Network
  - Core Network Policy
  - Segments
  - Network Function Groups
  - Visualization



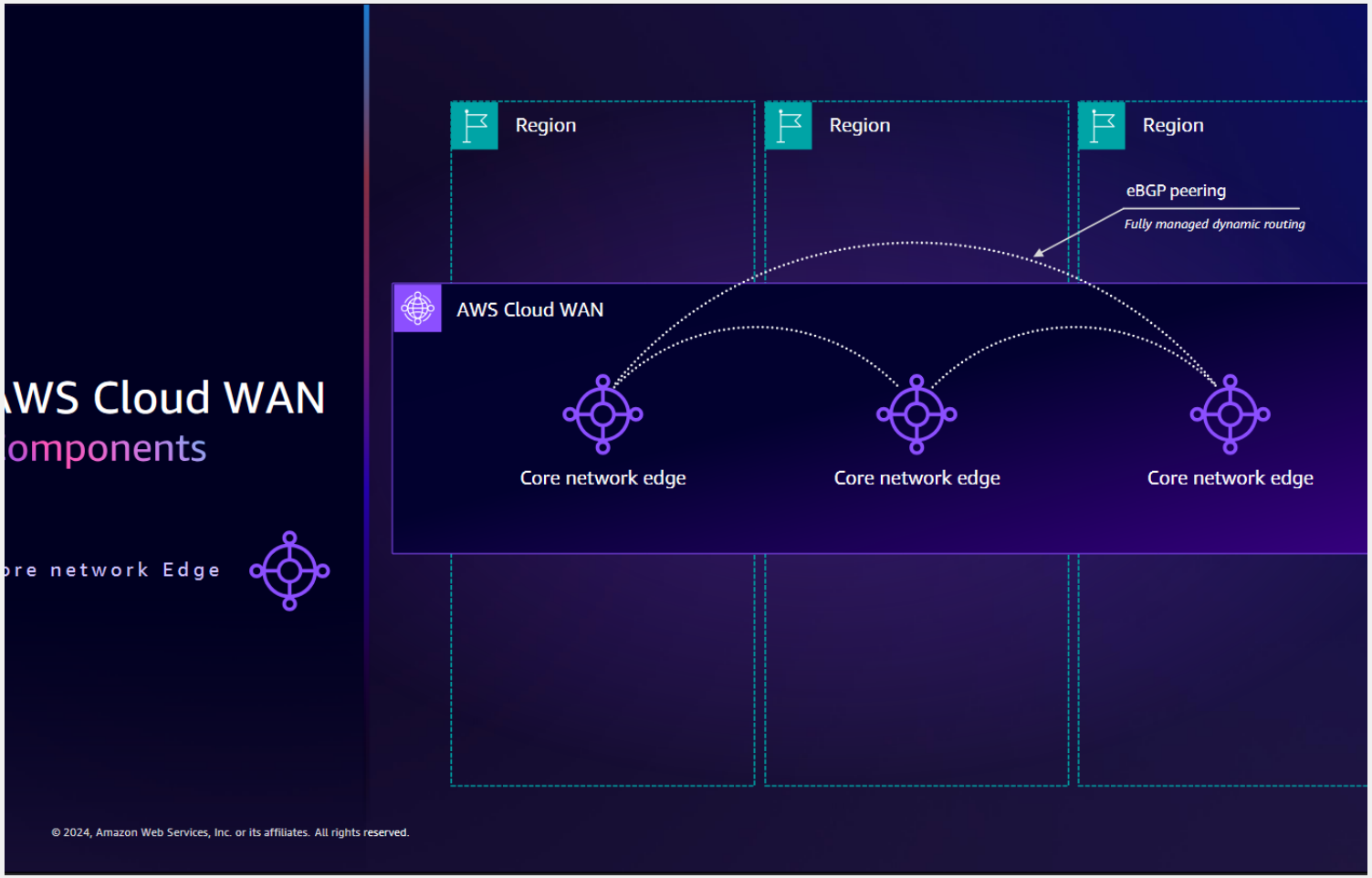








AWS Cloud WAN



Core Network

- Managed by AWS, and defined at creation

Core Network Edge

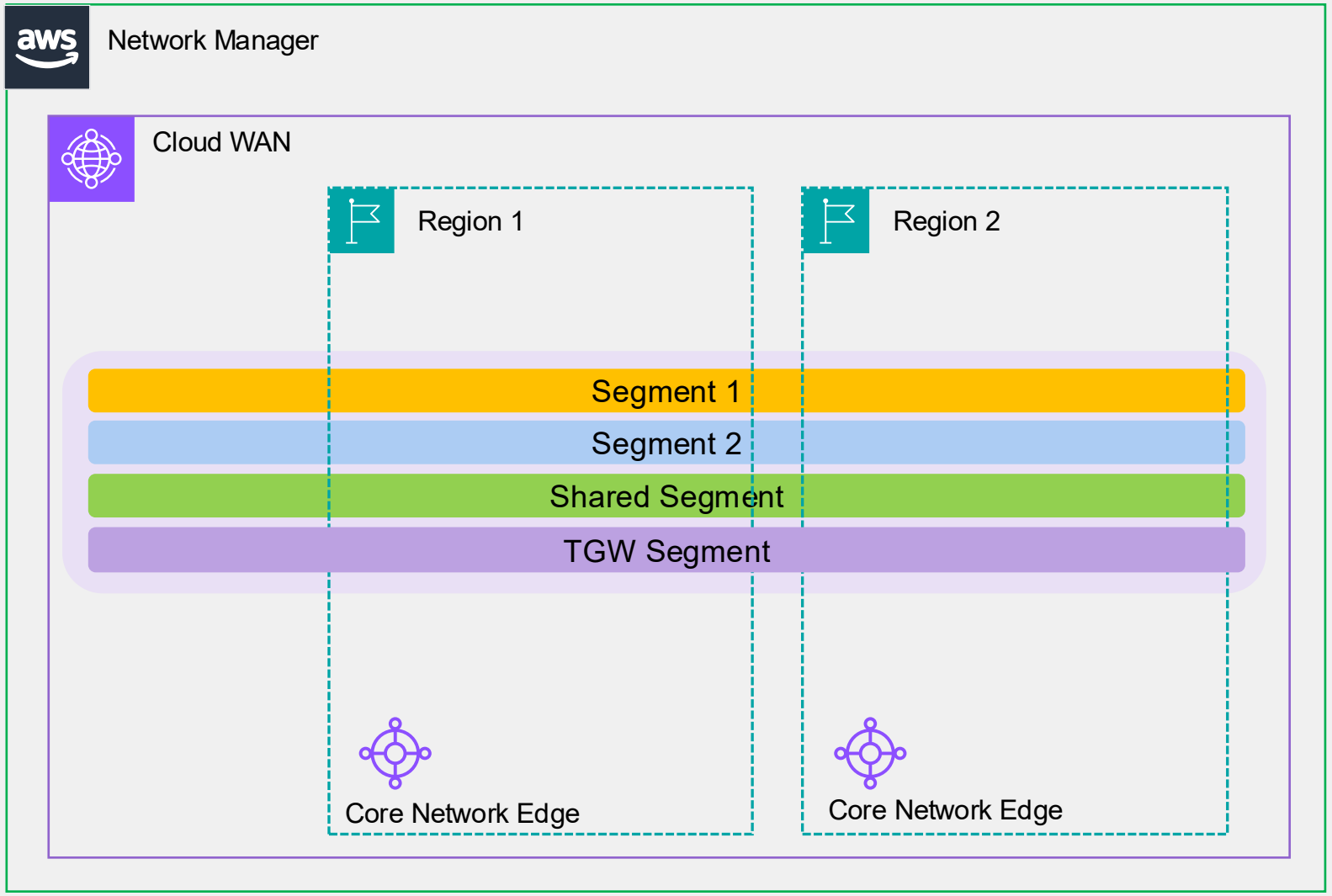
- Every CloudWAN attachment connects to a Core Network Edge in the region it exists
- Uses BGP for automated network learning

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AWS Cloud WAN

Key Components:

- Segments
  - Global by default
  - Acceptance – VPC self attach
  - Isolated
    - Specific permissions
    - Static Routes
  - Segment Filter
    - Filters shared routes learned from other segments

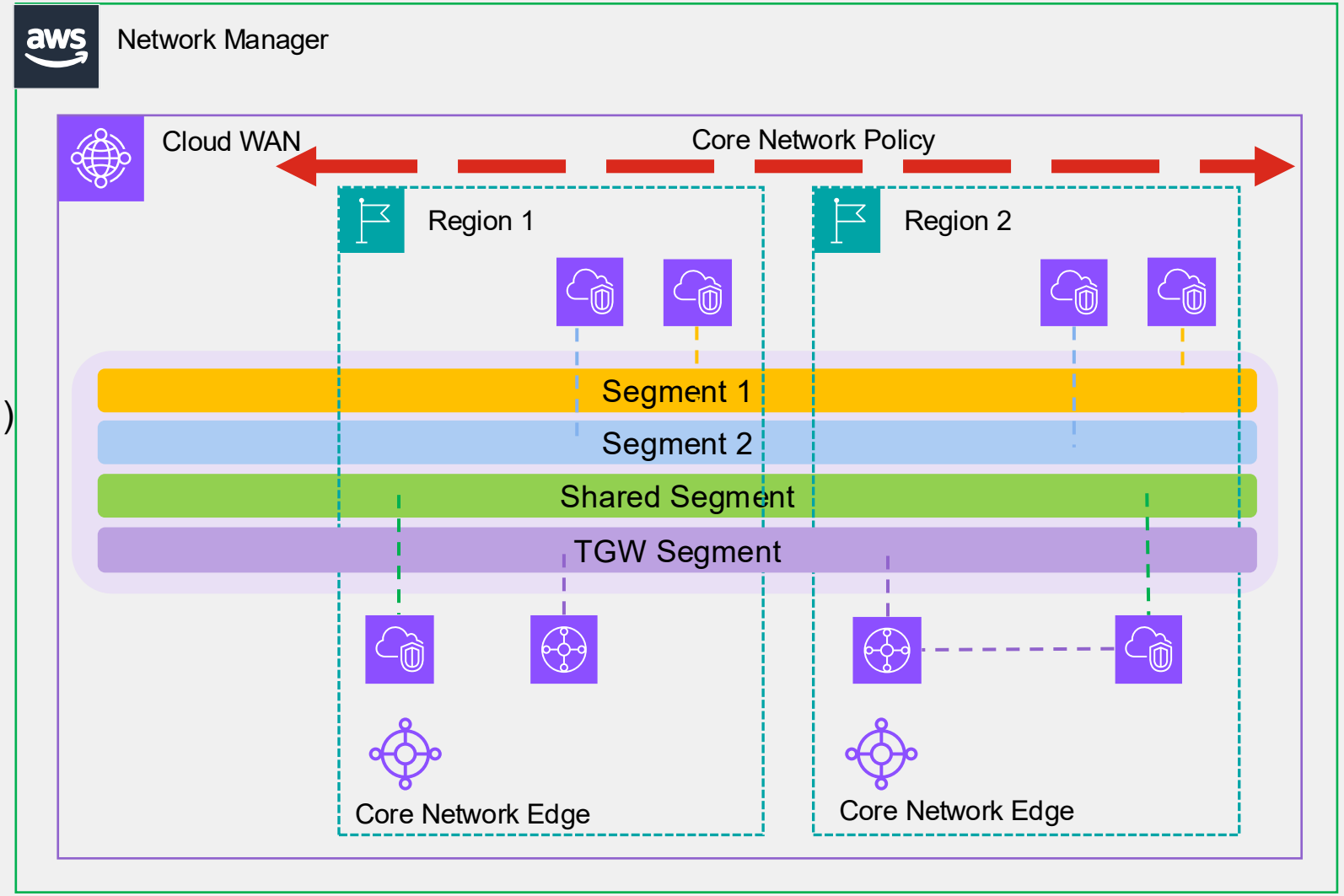


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AWS Cloud WAN

Key Components:

- Core Network Policy
  - JSON based
  - Version Control
  - Policy Types
    - Core Network (region/ASN)
    - Segment
    - Segment Actions
    - Attachment Policies

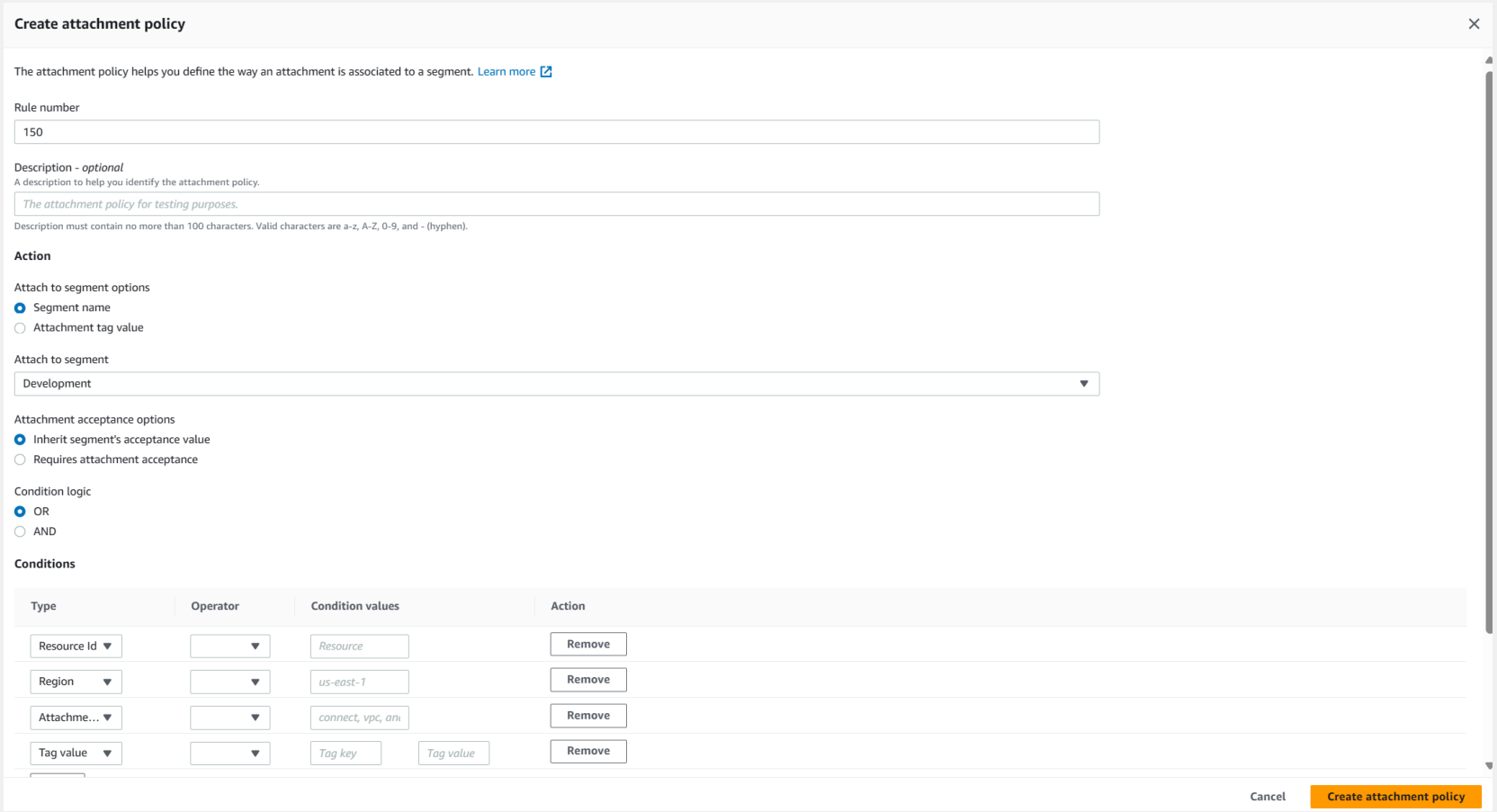


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AWS Cloud WAN

Key Components:

- Attachment Policy
  - Attachment Automation
  - Policy id prioritization
    - Tagging/Meta-data
  - Routing Simplification
    - Per region per TGW per attachment route tables vs
    - Per Segment

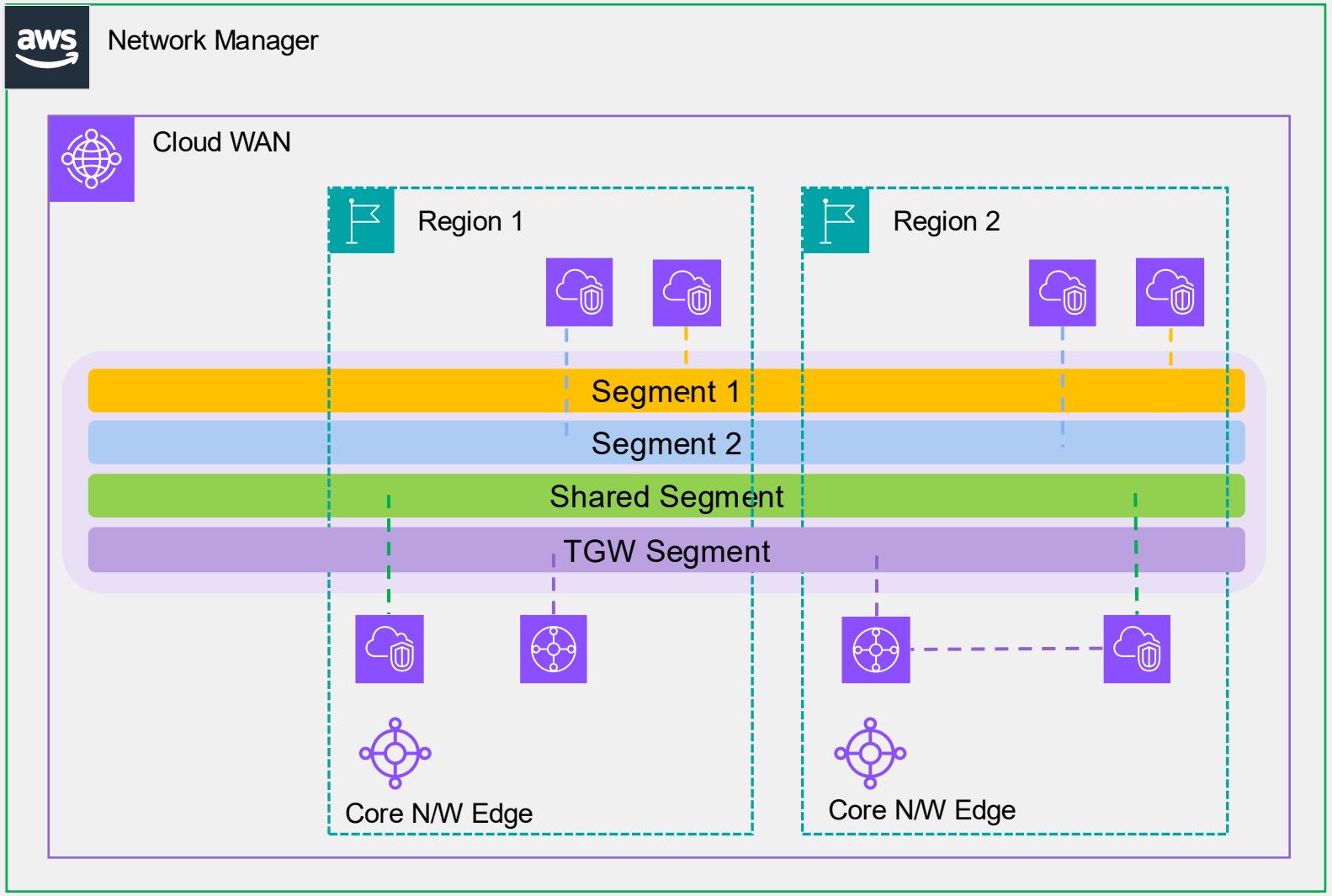


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AWS Cloud WAN

Key Components:

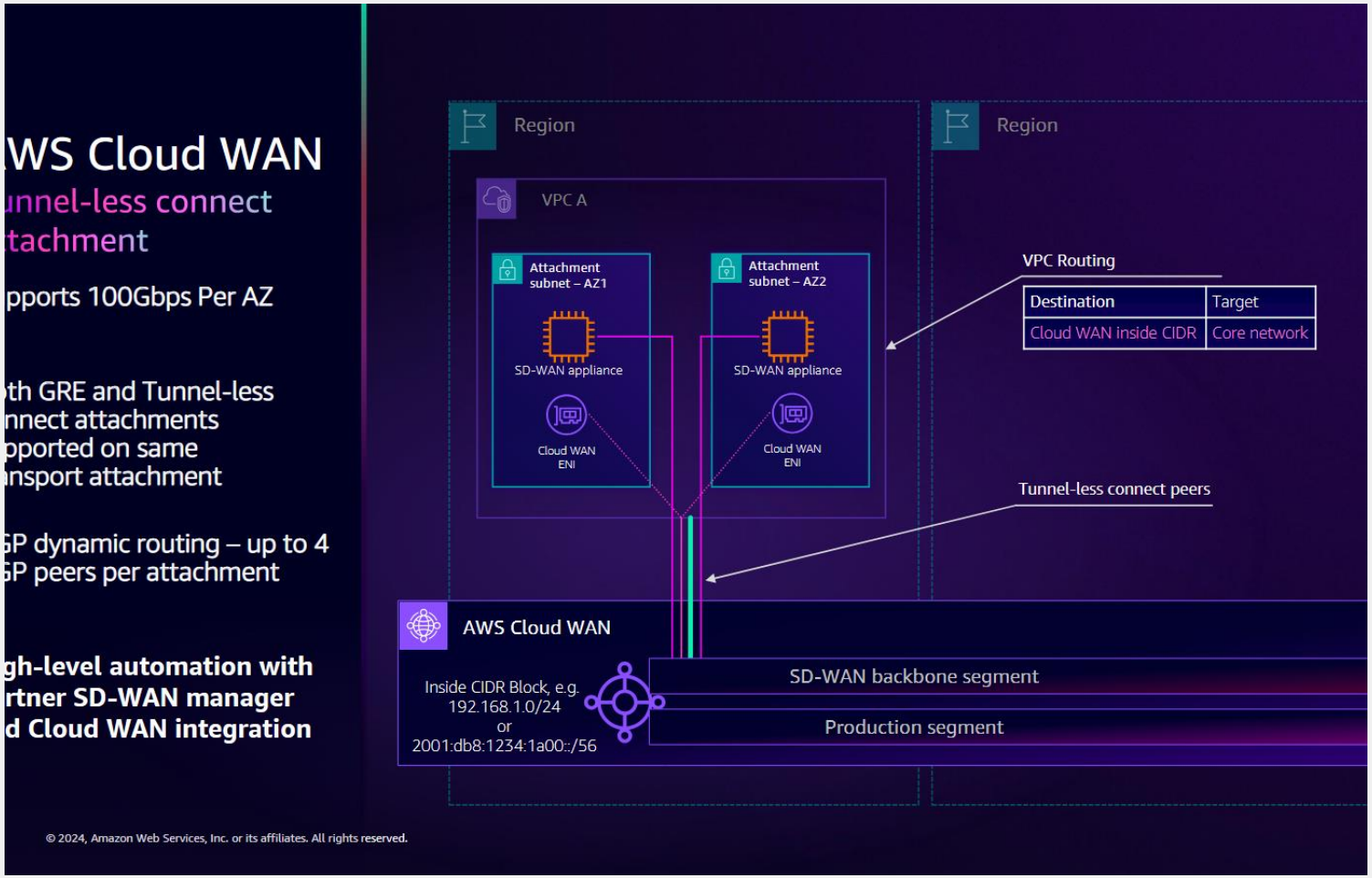
- Attachments
  - VPC
  - VPN
  - Direct-Connect
  - Connect*
  - Transit Gateway
  - Network Function Group



*IPSec, GRE, or Tunnel-less Connect available

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Tunnel-less connect



CloudWAN introduces tunnel-less connect attachments

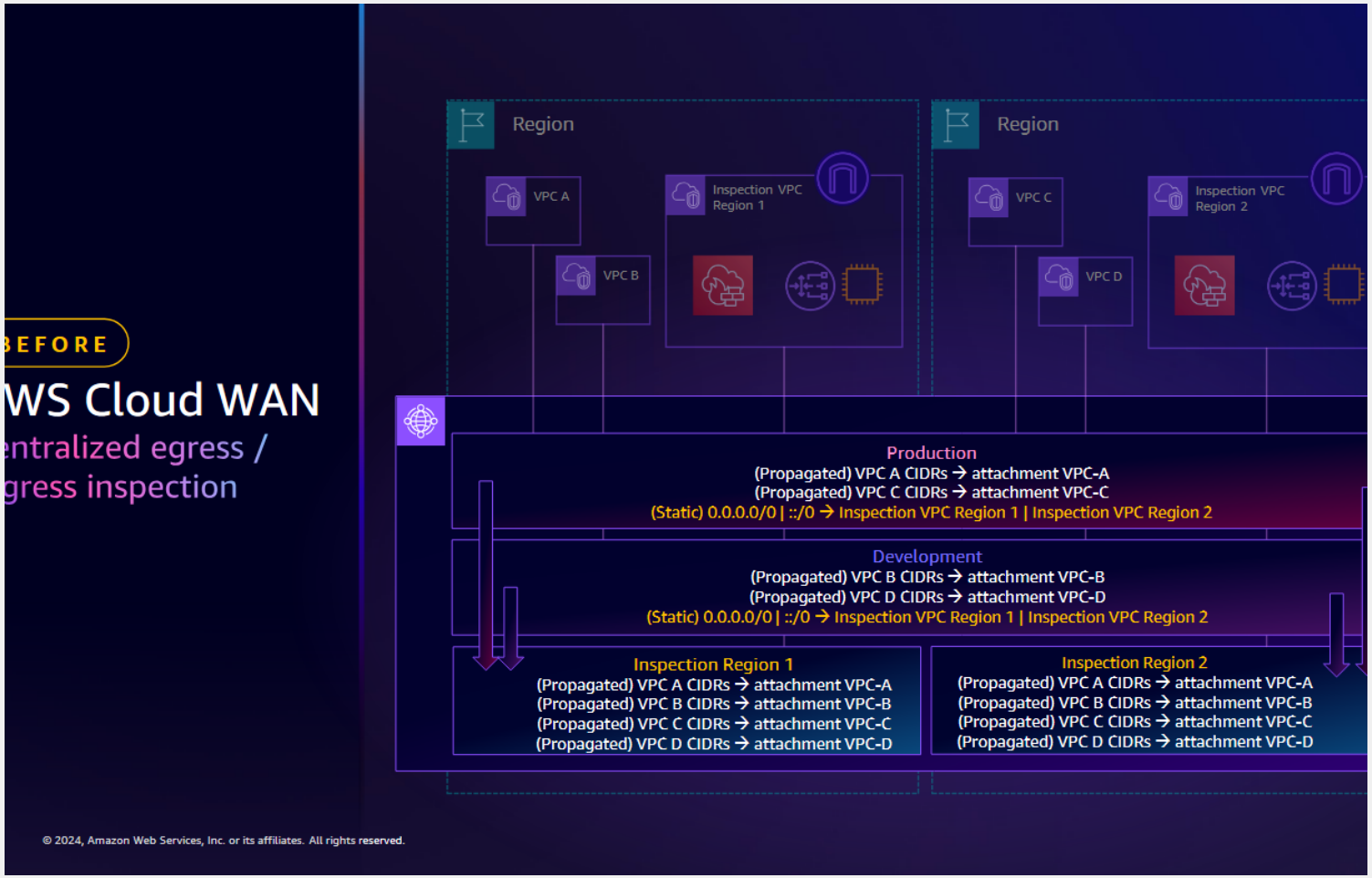
These attachments allow for BGP peering without the overhead of an IPSec or GRE tunnel

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What does it look like?

Cloud WAN Routing

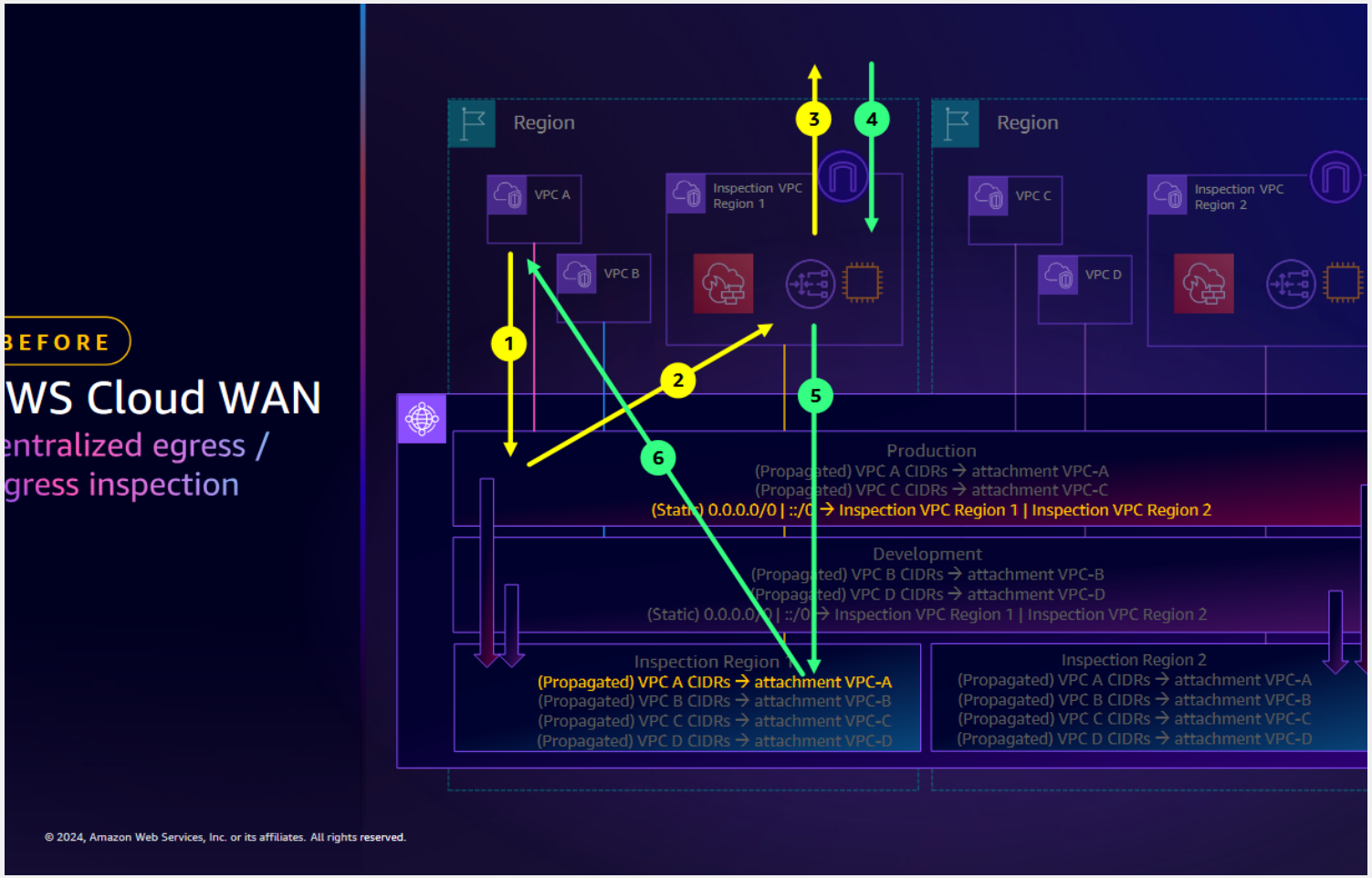
Per Segment Routing



All routing is propagated, no manual configuration needed

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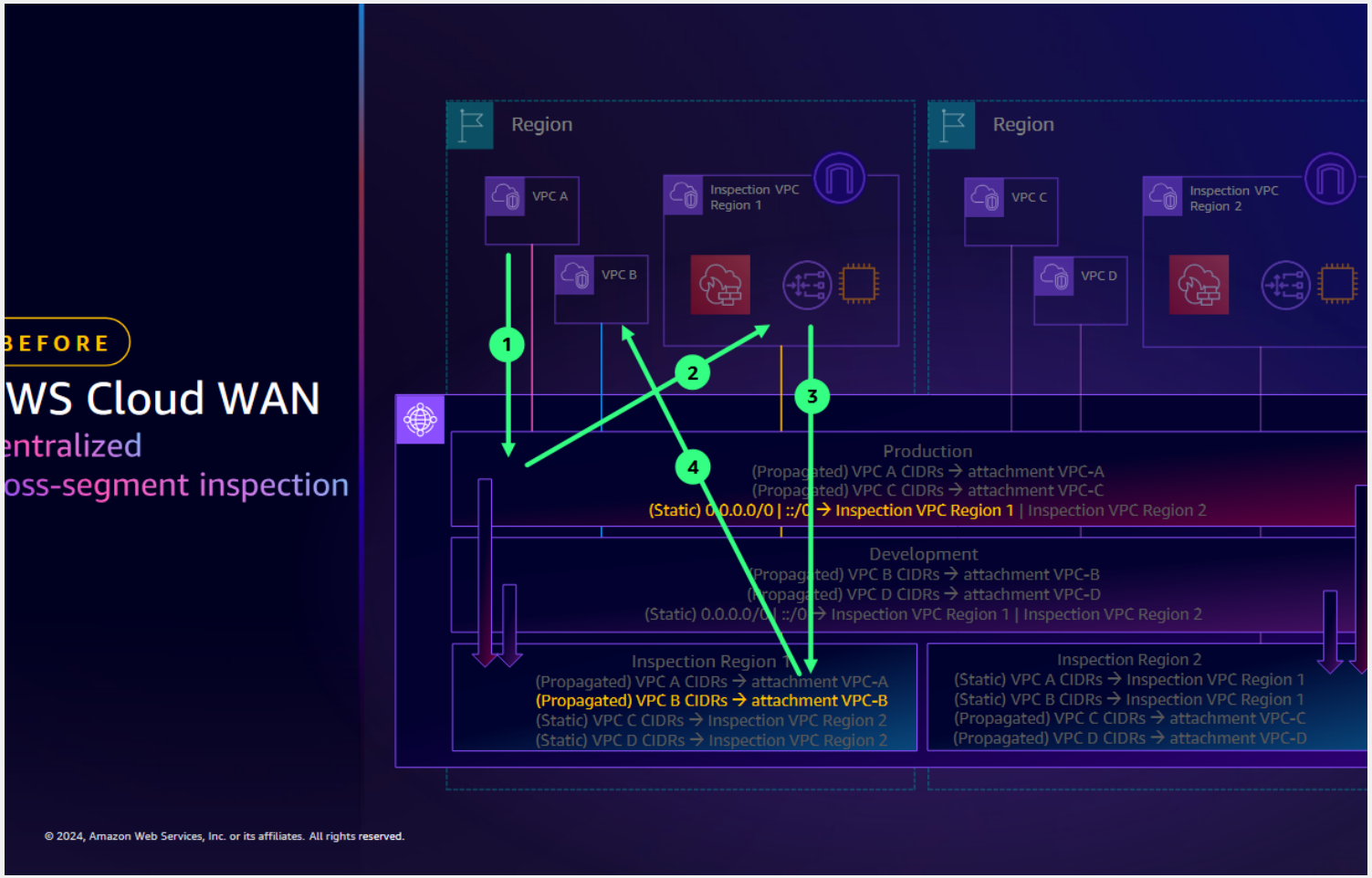
North-South Routing



Shows segment route tables and route path for north-south inspection

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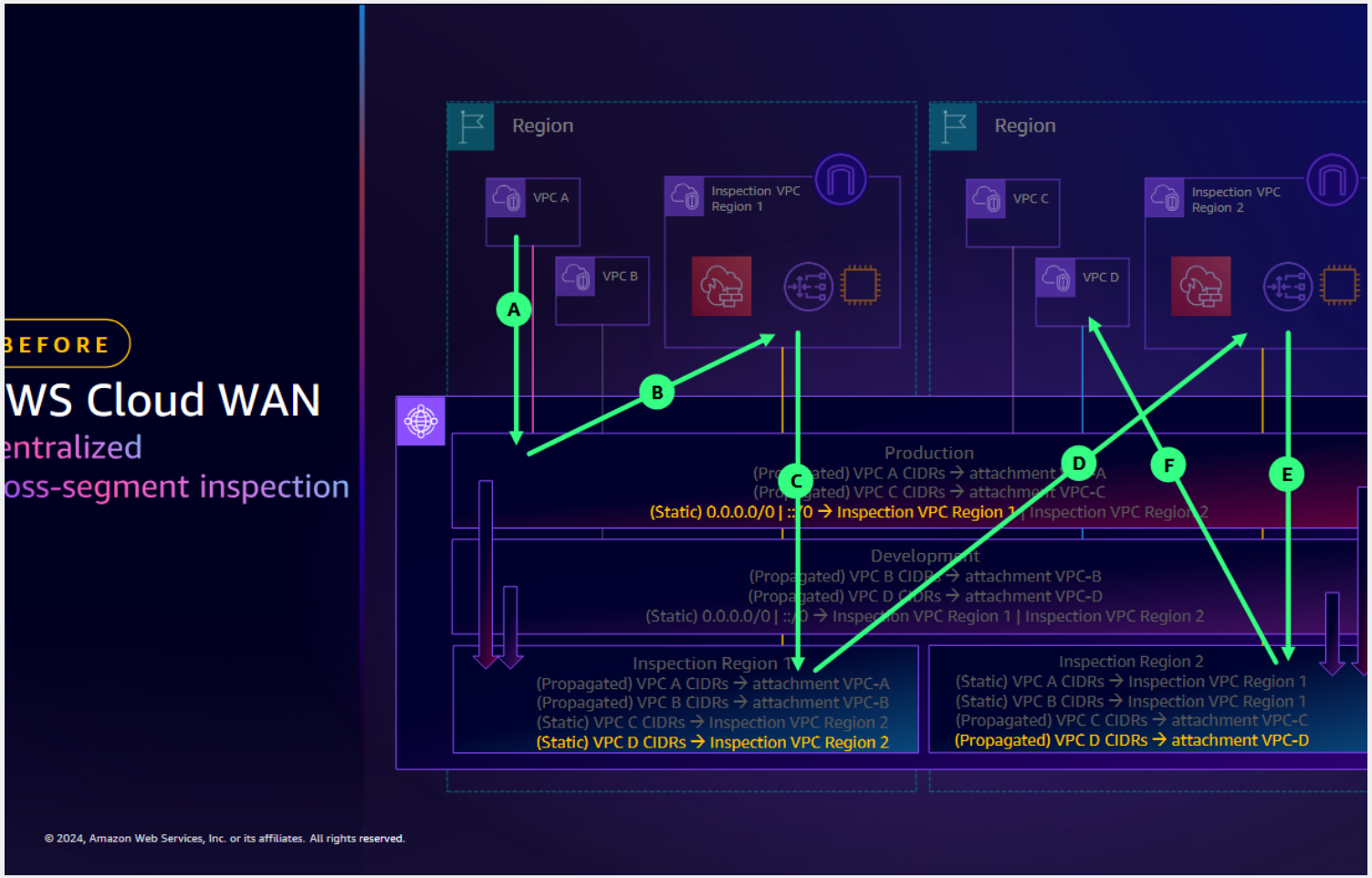
East-West Routing



Shows segment route tables and route path for east-west inspection

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Inter-region East-West

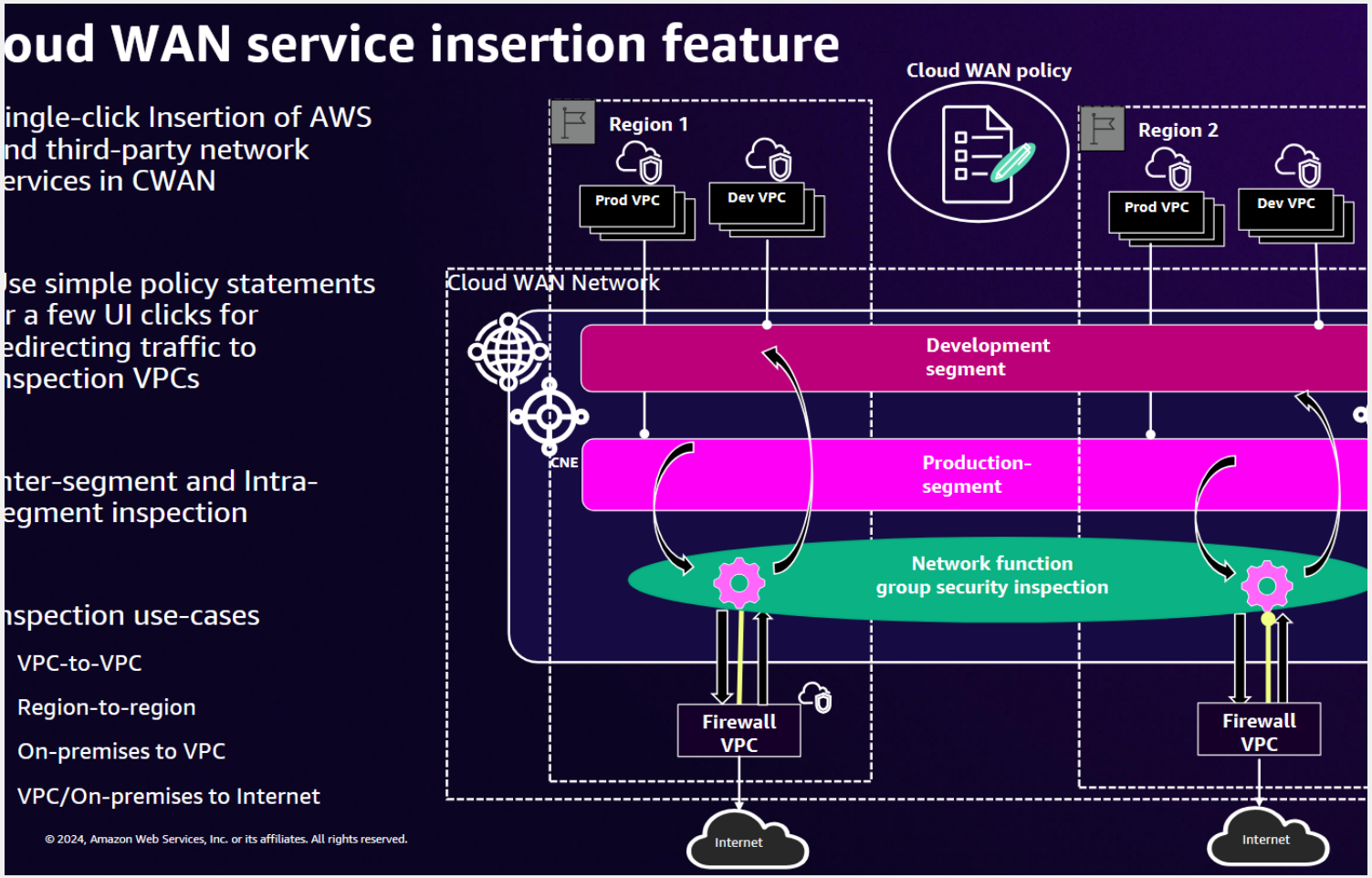


Routing path for inter-region east-west inspection

Dual-hop inspection is used to ensure all traffic is inspected.

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Service-Insertion (Network Function Group)



Newly announced at Re:Invent 2024 is the Network function group.

Typically integrated with Gateway Load Balancer, for scalability.

Network function groups are specially designated segments, with which you can declare an inspection via Core Network Policy (send-via or send-to action)

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Network Policy Example

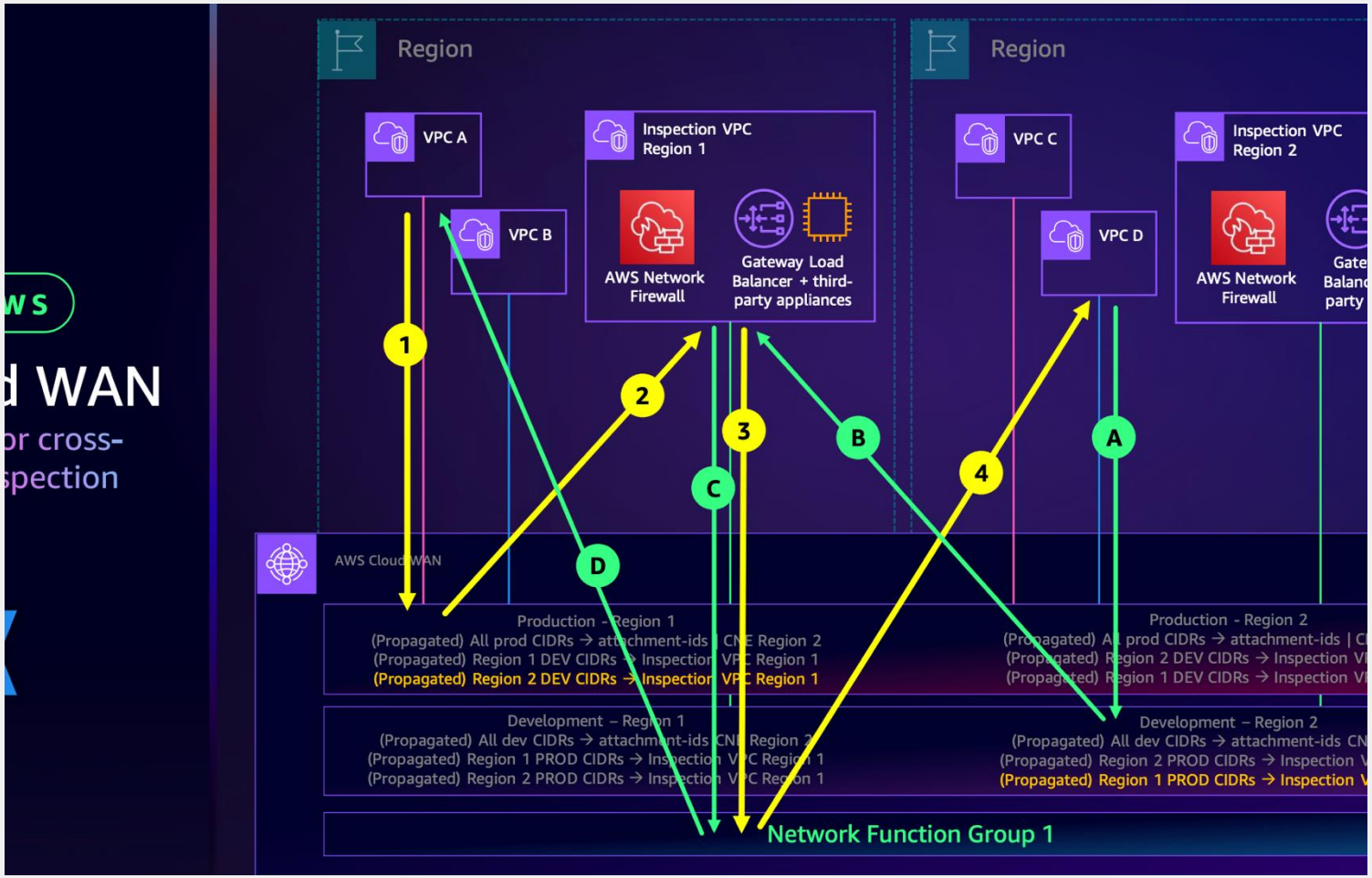


Segment policy to send all traffic to NetworkFunctionGroup1 when destination is ‘Development’ segment.

Optional edge overrides configured to use us-east-1

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E/W inter-region with NFG (send-via single hop)

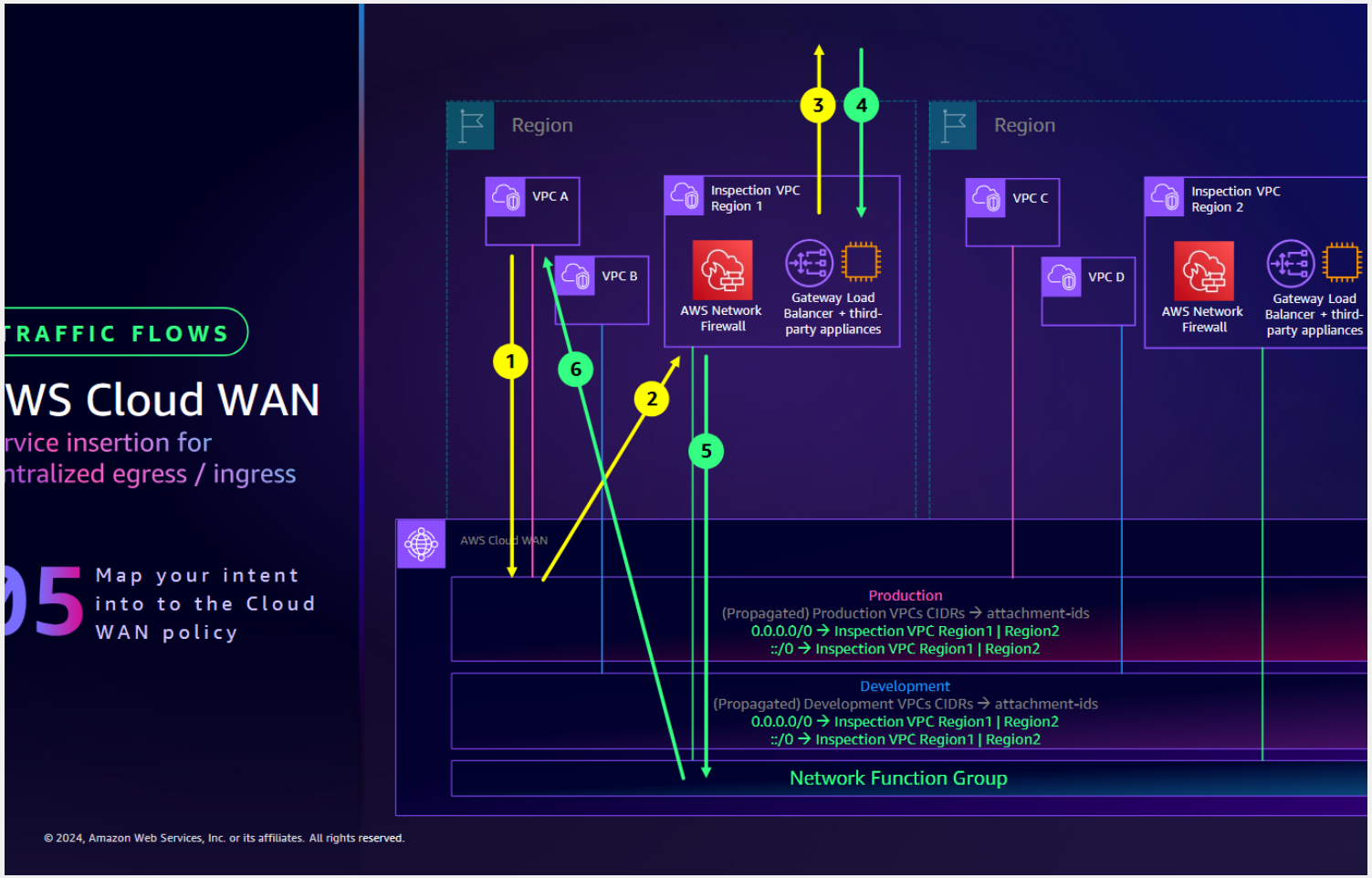


The routing tables have been greatly simplified with the NFG mappings in CloudWAN policy, and traffic remains inspected as before.

If a region doesn't have an NFG attachment, CloudWAN will automatically select an egress region based on region priorities

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Egress with NFG (send-to)

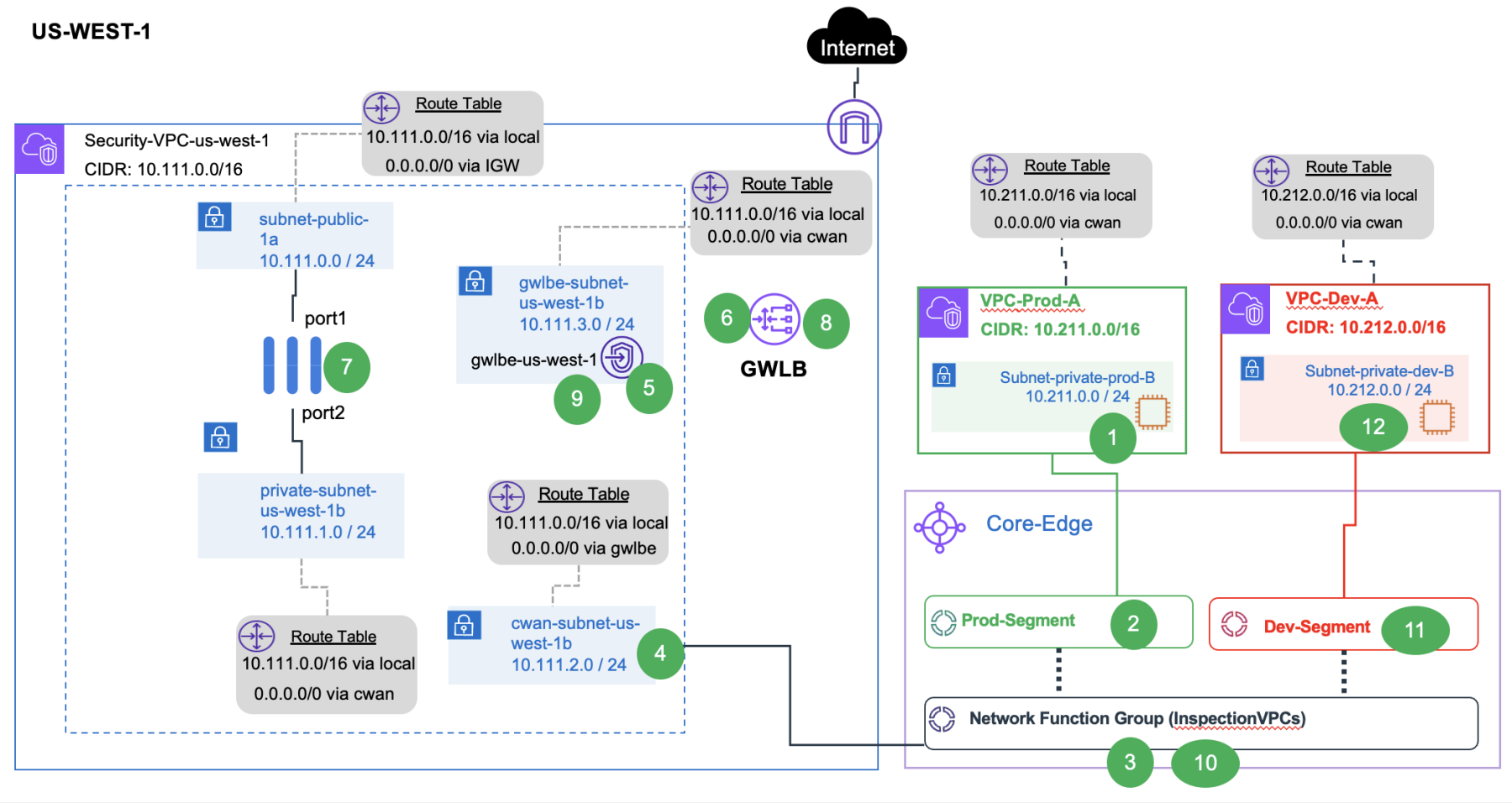


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FortiGate Service Insertion Diagram



Hop by hop diagram of service insertion from Fortinet's perspective.

The Security VPC is attached to an NFG.

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NFG & Service Insertion Key Takeaways

- An NFG is global like a segment
- Only one attachment per NFG per region
- Service insertion rules are bi-directional, they handle return traffic symmetrically. However, if both segments need to initiate communication in both directions, then two rules are needed
- Manual static routes are not supported for NFGs by design. Use an inspection or SDWAN VPC associated to a regular segment instead

- Send-to is for egress only, ie 0.0.0.0 & ::0 to the NFG, can't specify specific CIDR
- Send-via is for east/west between segments, both for inter and intra-segment inspection
- Send-via single hop is for deployments where inspection is only needed in one region and not both for inter-region traffic
- Send-via dual hop is for deployments where inspection is needed in both regions for inter-region traffic
- NFGs used by send-via dual hop rules can't have send-via single-hop or send-to rules as well (known limitation)

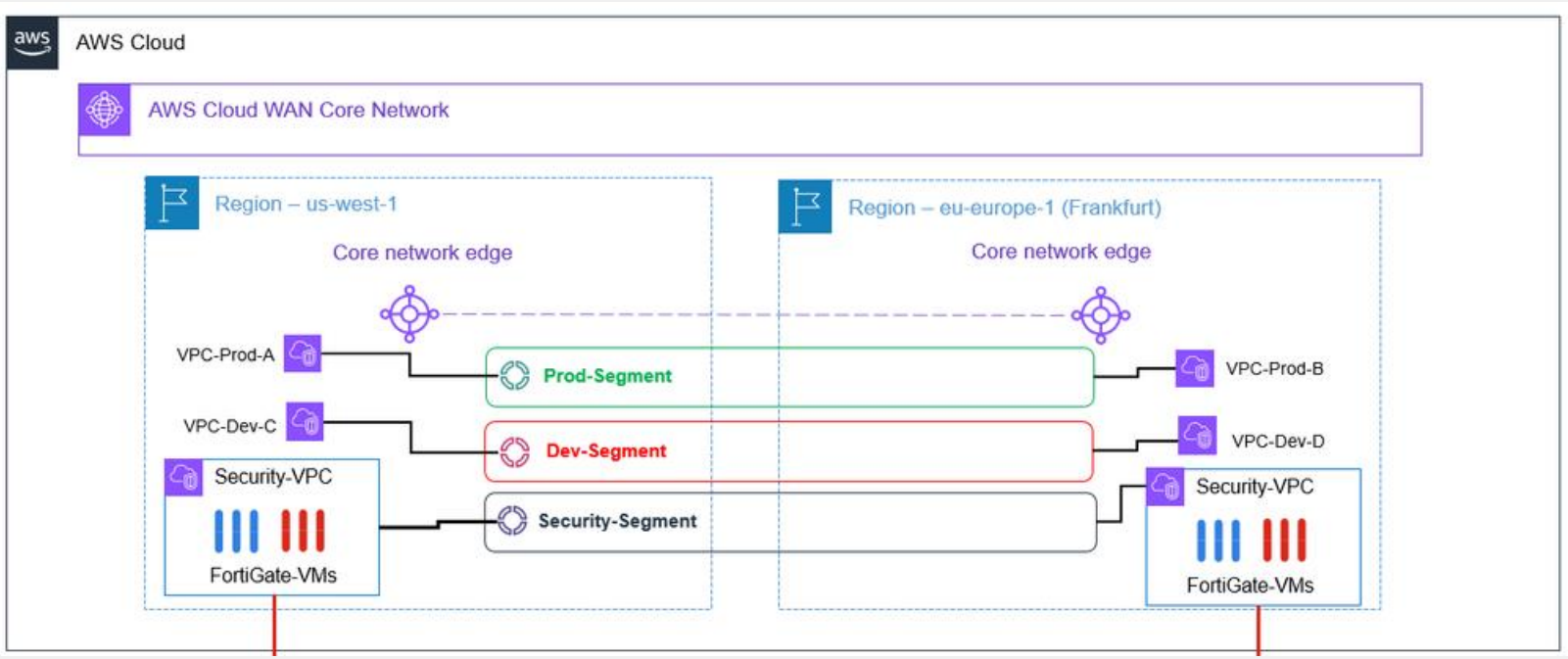
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Use Case 1: Globally Distributed App Security

Consistent Security & Network posture

Consistent Security and Network Posture

- Network Control
  - CM Network Team ownership
  - Simplified routing control
  - Per Segment globally
  - Automation of attachment
    - Use CI/CD or AWS landing Zone to force compliance
- Security Control
  - Segment/Isolate environments
    - Cross region consistency
    - AWS Well Architected framework Security Pillar
  - FOS Infra modelling approach
  - Cross Region Failover simplification
  - Ingress/Egress/E-W



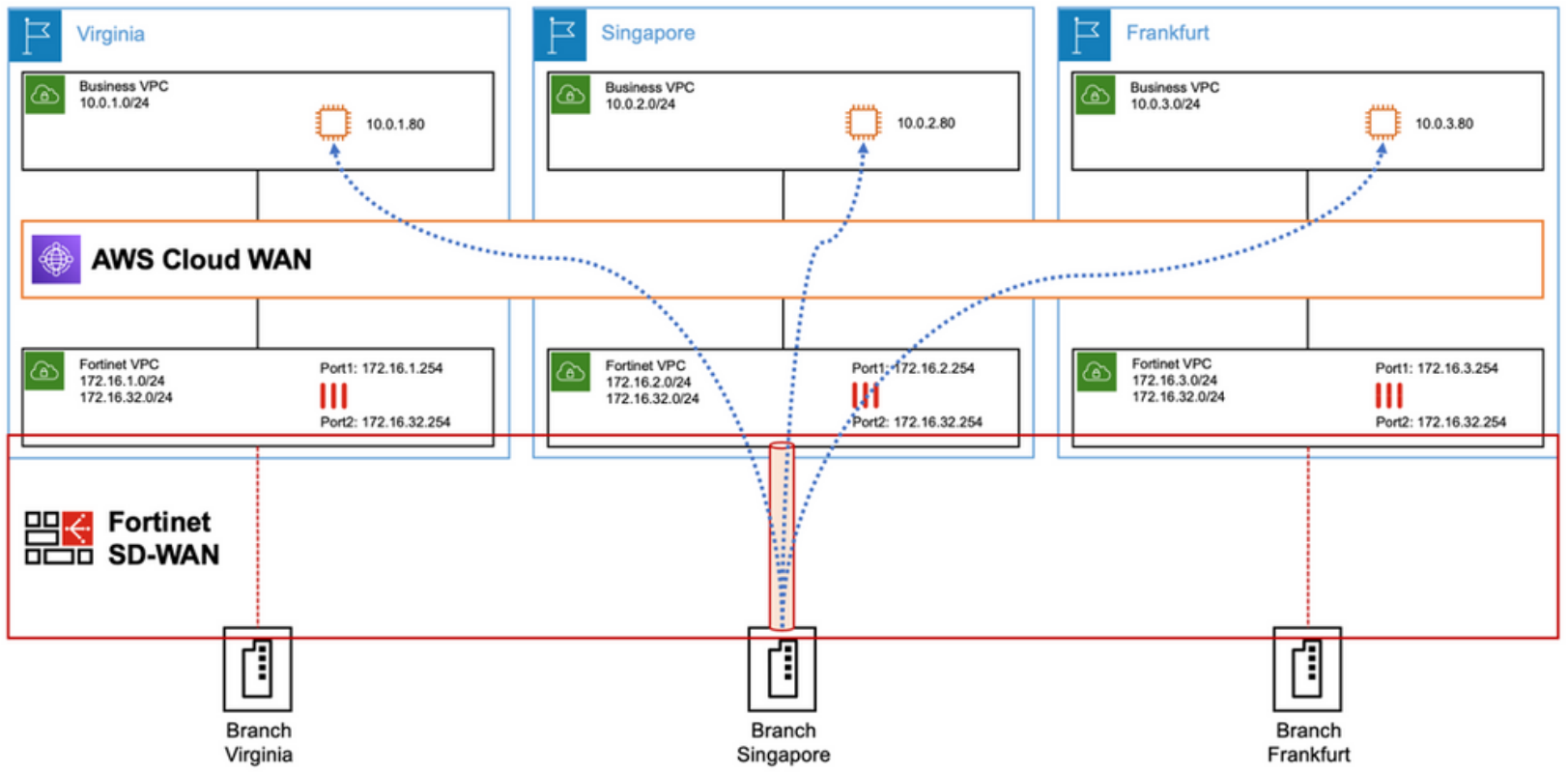
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Use Case 2: Optimised App Access with Fortinet SDWAN & AWS Cloud WAN

Improved Business Continuity and User Experience

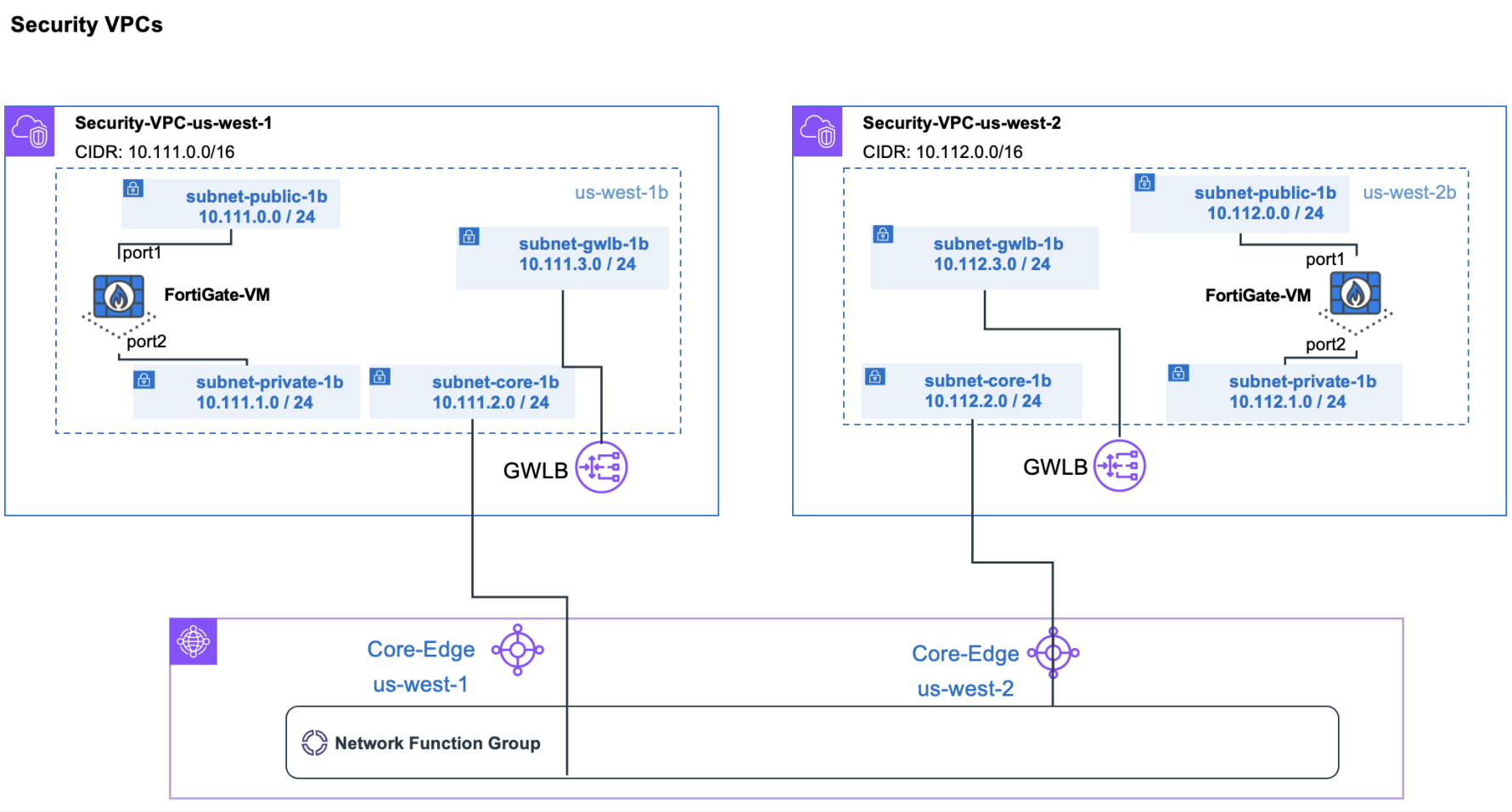
Faster App Access with Cross Region App deployment

- Cross Region App Access
  - Lower latency
  - More resilient
    - Fortinet SDWAN fabric cross region
    - Centralised Routing cross region



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Use Case 3: Service insertion with CloudWAN



- Network Function Group streamlines security inspection
- Can be used with FortiGate
- Can provide inspection in any direction

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Key Takeaways

- AWS CloudWAN unifies global cloud (AWS) network management across locations and AWS Regions.
- CloudWAN allows policy based management of security and traffic configurations
- Enhances connectivity options, with dynamic routing, and scalable throughput

- Fortinet architectures remain largely unchanged
- FortiGates can be added to a segment (SD-WAN tie-in), or a Network Function Group (Inspection)
- Since FortiGate has all the features on a single appliance, inspection can still be done in an ‘SD-WAN’ architecture.
- Nearly all CloudWAN integration is done via Core Network Policy.
- AWS has documentation around migrating to CloudWAN with Service Insertion, for both existing CloudWAN and Transit-Gateway customers.

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Resources

- **AWS CloudWAN re:Inforce 2024 talk:** https://reinforce.awsevents.com/content/dam/reinforce/2024/slides/NIS203-NEW_AWS-Cloud-WAN-service-insertion-Simplifying-network-inspection.pdf

  https://www.youtube.com/watch?v=hsIvkdtkBZE
- **Service Insertion walkthrough:** https://community.fortinet.com/t5/FortiGate-VM-on-AWS-Discussions/AWS-Cloud-WAN-Service-Insertion-with-FortiGate/m-p/357490
- **Optimized App Access Use case walkthrough:** https://community.fortinet.com/t5/Blogs/AWS-Cloud-WAN-integration-Fortinet-SD-WAN-enables-cloud-network/ba-p/238225
- **Secure SD-WAN walkthrough:** https://community.fortinet.com/t5/Customer-Service/Technical-Tip-AWS-Cloud-WAN-unleashed-with-Fortinet-SD-WAN/ta-p/280125

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